

Problem 49. Find all distinct prime numbers p, q and r such that

$$3p^4 - 5q^4 - 4r^2 = 26.$$

Solution. First notice that if both primes q and r differ from 3, then $q^2 \equiv r^2 \equiv 1 \pmod{3}$, hence the left hand side of the given equation is congruent to zero modulo 3, which is impossible since 26 is not divisible by 3. Thus, $q = 3$ or $r = 3$. We consider two cases.

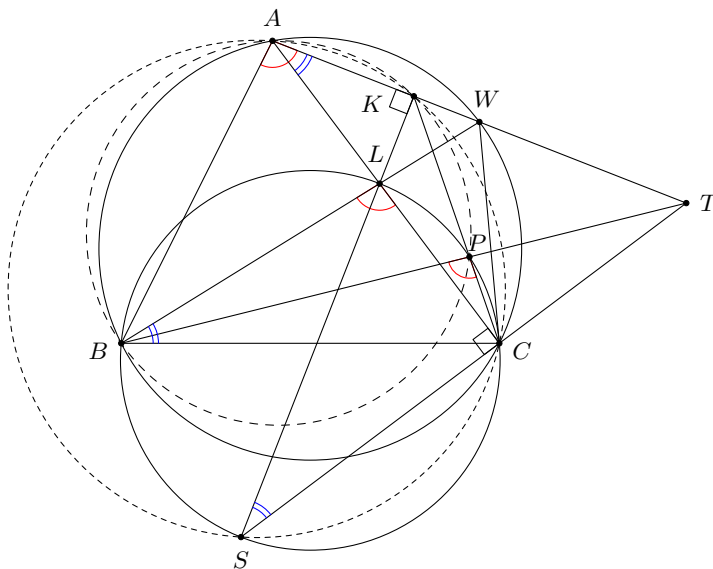
Case 1. $q = 3$.

The equation reduces to $3p^4 - 4r^2 = 431$. If $p \neq 5$, by Fermat's little theorem, $p^4 \equiv 1 \pmod{5}$, which yields $3 - 4r^2 \equiv 1 \pmod{5}$, or equivalently, $r^2 + 2 \equiv 0 \pmod{5}$. The last congruence is impossible in view of the fact that a residue of a square of a positive integer belongs to the set $\{0, 1, 4\}$. Therefore, $p = 5$ and $r = 19$.

Case 2. $r = 3$.

The equation becomes $3p^4 - 5q^4 = 62$. Obviously $p \neq 5$. Hence, Fermat's little theorem gives $p^4 \equiv 1 \pmod{5}$. But then $5q^4 \equiv 1 \pmod{5}$, which is impossible. Hence, the only solution of the given equation is $p = 5, q = 3, r = 19$. $\square$

Problem 50. Let ABC be a triangle. The internal bisector of $\angle ABC$ intersects the side AC at L and the circumcircle of triangle ABC again at $W \neq B$. Let K be the perpendicular projection of L onto AW . The circumcircle of triangle BLC intersects line CK again at $P \neq C$. Lines BP and AW meet at point T . Prove that $AW = WT$.



Solution 1. Note that

$$\angle BPC = \angle BLC = \angle BAC + \frac{\angle CBA}{2} = \angle BAW.$$

Therefore, $ABPK$ is cyclic. Let KL intersect the circumcircle of triangle BLC at a point $S \neq L$. Then

$$\angle KSC = \angle LSC = \angle LBC = \angle WBC = \angle WAC = \angle KAC.$$

So $KASC$ is cyclic.

Now, radical axes of circles $KABP, BPCS, KASC$ are concurrent. That is: AK, BP, CS concur and so CS passes through T . Note that $\angle SCA = \angle SKA = 90^\circ$, so $\angle TCA = 90^\circ$. As $AW = WC$, we find that W is the circumcentre of the right triangle ACT , so $AW = WT$. $\square$

Solution 2. Let T' be the point on the ray AW such that $AW = WT'$. We will show that $T' = T$.

As $WA = WC = WT'$, C lies on the circle with diameter AT' , so $\angle ACT' = 90^\circ = \angle LKT'$. It follows that $KLCT'$ is cyclic.

Note that

$$\angle ALW = \angle BLC = \angle BAC + \frac{\angle CBA}{2} = \angle BAW.$$

consecutive). In other words, the difference of these two fractions is less than $\frac{1}{2}$. As the indices involved in these two fractions are distinct, we can choose them to be i, j, k, ℓ and conclude that $C \leq \frac{1}{2}$.

Next, we show that $C = \frac{1}{2}$ is best possible. Consider the numbers $1, 2, 2, 2, n$ where n is a large real number. The fractions formed by two of these numbers in ascending order are $\frac{1}{n}, \frac{2}{n}, \frac{1}{2}, \frac{2}{2}, \frac{2}{1}, \frac{n}{2}, \frac{n}{1}$. Since the indices i, j, k, ℓ are distinct, $\frac{1}{n}$ and $\frac{2}{n}$ cannot be chosen simultaneously. Therefore the minimum value of the left-hand side of (1) is $\frac{1}{2} - \frac{2}{n}$. When n tends to infinity, this value approaches $\frac{1}{2}$, and so C cannot be less than $\frac{1}{2}$.

These conclude that $C = \frac{1}{2}$ is the smallest possible choice. □